

ANSWER KEY — Integration Lesson 8 Test: FTC Practice, Polynomial Fluency

For the grader · full working shown · give credit for correct method with small arithmetic slips · fraction answers must be exact (equivalent forms OK)

1. Evaluate $\int_0^4 x \, dx$.

Work: $F(x) = x^2/2$ (check: derivative is x ✓). $F(4) = 16/2 = 8$; $F(0) = 0$. Answer: $8 - 0$. **Answer: 8.**

2. Evaluate $\int_0^2 x^2 \, dx$.

Work: $F(x) = x^3/3$ (check: derivative is x^2 ✓). $F(2) = 8/3$; $F(0) = 0$. **Answer: 8/3.** (Classic error: $x^3/2$ — dividing by the OLD power — gives 4, wrong.)

3. Evaluate $\int_0^5 3 \, dx$, then check with a rectangle picture.

Work: $F(x) = 3x$. $F(5) - F(0) = 15 - 0 = 15$. Picture check: region under $y = 3$ from 0 to 5 is a rectangle, width $5 \times$ height $3 = 15$ ✓. **Answer: 15.**

4. Evaluate $\int_0^2 (2x + 5) \, dx$.

Work: term by term: $2x \rightarrow x^2$, $5 \rightarrow 5x$, so $F(x) = x^2 + 5x$ (check: derivative is $2x + 5$ ✓). $F(2) = 4 + 10 = 14$; $F(0) = 0$. **Answer: 14.** (Classic error: dropping the constant term — forgetting $5 \rightarrow 5x$ gives just 4, wrong.)

5. Evaluate $\int_0^1 x^3 \, dx$.

Work: $F(x) = x^4/4$ — raise to power 4, divide by the NEW power 4 (check: derivative is $4x^3/4 = x^3$ ✓). $F(1) = 1/4$; $F(0) = 0$. **Answer: 1/4.** (Classic error: $x^4/3 \rightarrow 1/3$, wrong.)

6. Evaluate $\int_1^2 x^2 \, dx$.

Work: $F(x) = x^3/3$. $F(2) = 8/3$; $F(1) = 1/3$. Subtract: $8/3 - 1/3 = 7/3$. **Answer: 7/3.** (Classic error: forgetting to subtract $F(1)$ gives $8/3$, wrong — the bottom limit isn't 0 here.)

7. Evaluate $\int_1^3 (2x + 1) \, dx$.

Work: $F(x) = x^2 + x$ (check: derivative is $2x + 1$ ✓). $F(3) = 9 + 3 = 12$; $F(1) = 1 + 1 = 2$. Subtract: $12 - (2) = 10$. **Answer: 10.**

8. Evaluate $\int_0^1 (6x^2 + 2x) dx$.

Work: $6x^2 \rightarrow 6 \cdot x^3/3 = 2x^3$; $2x \rightarrow x^2$. $F(x) = 2x^3 + x^2$ (check: derivative is $6x^2 + 2x$ ✓). $F(1) = 2 + 1 = 3$; $F(0) = 0$. **Answer: 3.**

9. Evaluate $\int_1^2 (3x^2 - 2x) dx$.

Work: $F(x) = x^3 - x^2$ (check: derivative is $3x^2 - 2x$ ✓). $F(2) = 8 - 4 = 4$; $F(1) = 1 - 1 = 0$. Subtract: $4 - (0) = 4$. **Answer: 4.** ($F(1)$ happens to be 0, but students must still show the subtraction — the question asks for it. Writing $F(a)$ in parentheses is the habit being graded.)

10. Evaluate $\int_0^2 (x^3 - x + 2) dx$, with a written derivative check.

Work: $F(x) = x^4/4 - x^2/2 + 2x$. Required check: derivative is $4x^3/4 - 2x/2 + 2 = x^3 - x + 2$ ✓. $F(2) = 16/4 - 4/2 + 4 = 4 - 2 + 4 = 6$; $F(0) = 0$. **Answer: 6.** (Classic errors to watch: dropping the +2 term, or a sign slip on $-x^2/2$. Deduct half if the "check ✓" step is missing — it was part of the question.)